



Short Communication

Evaluation of heavy metal pollution from coastal water of Nizampatnam Bay and Lankevanidibba, East Coast of India

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ARTICLE INFO

Keywords:

Coastal water
Anthropogenic activities
Heavy metals
East Coast of India

ABSTRACT

Thirty locations in the Coastal area of Nizampatnam Bay and Lankevanidibba were studied for heavy metals and their concentrations in the Coastal water. The samples were analysed for Cd, Co, Cr, Fe, Mn, Pb, and Zn to evaluate metal concentrations in coastal water. The concentration of metals Cd, Co, Cr, Fe, Mn, Pb, and Zn were determined by ICP-OES, in coastal water. The concentration of Cd (Cadmium) is demonstrating elevated contamination in all the sampling sites. The coastal water is polluted by heavy metals like Mn and Zn. It acts as a sink for the multitude of anthropogenic sources in the study area. The metals were identified in all coastal water samples were proxies of pollution. Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-OES-Perki Elmer-Optima 4300 DV) is an analytical technique used for elemental determinations and instrument was utilized for water analysis. Based on the order of accumulation the metal content was more and as a result pollution was more in the coastal water due to release of industrial wastes and anthropogenic activities. The comparison of Premonsoon and post monsoon is given.

Premonsoon: Pb (16.049 µg/L), Zn (7.998 µg/L), Mn (15.976 µg/L), Ni (8.827 µg/L), Fe (10.028 µg/L), Cd (9.221 µg/L), Cu (11.744 µg/L), Cr (10.486 µg/L), post monsoon: Pb (15.952 µg/L), Zn (8.084 µg/L), Mn (20.437 µg/L), Ni (8.583 µg/L), Fe (0.763 µg/L), Cd (4.837 µg/L), Cu (12.116 µg/L), Cr (8.008 µg/L).

1. Introduction

The world is very much an important part of human life. It is caused by the interaction of natural sources such as air, atmosphere, biosphere, Earth, and water with human-produced components (Visan et al., 1998; Abouhend and El-Moselhy, 2015). All of these engage with influence the existential dynamics and possible growth possibilities of society. Demarcation of protected ecosystems is warranted in order to preserve them from further decay and determination. It is important to protect the environment's quality by reducing the lethal effects caused by anthropogenic activity resources. Potentially toxic metals emitted by anthropogenic activities trigger significant ecosystem disruption (Ansari et al., 2004; Maha et al., 2008). Water contamination is a dynamic mechanism that changes the aquatic flora and fauna's water

composition. This may result in a shady form for economic and recreational purposes that is harmful to human health (Novotny, 2003). The importance of elevated metal values in natural waters and the atmosphere for pollution monitoring studies is lethal. Traces of metal ions play a crucial role in a range of biological activity. Some of these heavy metal levels are extremely high, such as Fe, Ni and Pb, which have a negative impact on the central nervous system. Since heavy metals are found in very small amounts in nature, their presence is normally not harmful to the environment. However, it is dangerous to humans if present in significant amounts (which is generally due to industrial activities). Heavy metal contamination of the marine ecosystem has become a worldwide problem in recent years. Toxic heavy metal pollutants are produced directly from atmospheric deposition, geologic weathering, or industrial waste effluents. Because of their possible

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<https://doi.org/10.1016/j.seares.2022.102232>

Received 4 January 2022; Received in revised form 23 May 2022; Accepted 28 May 2022

Available online 3 June 2022

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toxicity and capacity to bioaccumulation in marine environments, research into the degree and distribution of contamination of heavy metals in the fragile coastal region has recently piqued the public interest (Martorell et al., 2009). Metals such as Cd, Cr, Ni, and Pb are often regarded as markers of anthropogenic influence in the marine environment as a whole, as well as a possible danger to the natural environment (Sreenivasulu et al., 2016).

Due to high heavy metal emission, several researchers have demonstrated the importance of evaluating metal distribution in marine surface sediment (Sreenivasulu et al., 2017). As a result, assessing and evaluating these heavy metals in the coastal environment is a critical challenge, particularly in areas where pollution levels are troubling, such as when local factories discharge untreated wastewater directly into the sea, causing significant harm to marine ecology and aquatic life, including humans (Passeri et al., 2015). Their impact is more visible when abnormal values of a series of physics chemical parameters are used. The importance of heavy elements in coastal water is rapidly becoming a global problem that must be tackled (Karbassi et al., 2018). As a result, both scientific and statutory agencies are concerned about the assessment of heavy metal emissions in the research region. So far, little work has been done on heavy metals near this coastal region (Nouri et al., 2008). During the winter of 2016, thirty coastal water samples were collected from Nizampatnam Bay and Lankevanidibba for this research. The primary goals of this research are to evaluate the spatio-temporal distribution of heavy metal emissions in the field.

2. Study area

Earlier workers have studied the coastal Nizampatnam Bay, East Coast of India on several aspects such as geomorphic aspects, provenance (Nageswara Rao et al., 2008) environmental analysis. However, this paper attempts to study metal distribution in coastal water concerning coastal processes in Nizampatnam Bay and Lankevanidibba. (Jayaraju et al., 2009; Sreenivasulu et al., 2016) (Fig. 1).

2.1. Geology and geomorphic setting

The study area is covered with recent alluvium. The general geomorphological features include estuaries, mangrove swamps, modern beach dune ...etc. In the research area, at least five sets of palaeo sandy dunes were discovered. Before it debouches, the Krishna River splits into at least three branches, each about 10 km long: Gollumuttapaya, the main Krishna River, and Nadimeru. The main Krishna River today carries the majority of the runoff and reaches the Bay at False Divi. At the lower delta level, tidal channels are present. It is an embayment on its southern side, bordering the Krishna delta. It has a coastline of around 125 km between the SW point of Kottapatnam and the NE point of False Divi. The sediment sample sites receive large quantities of untreated effluents from industries areas, Sangu gas production plant and shipping activity.

3. Materials and methods

The purpose of this research is to focus on the quality of coastal waters utilized for human consumption in two localities in Nizampatnam Bay and Lankevanidibba. A total of 30 samples were collected from fixed sampling profiles of the study area. The Water samples were collected in high density previously washed polyethylene containers (Ramkumar and Ramayya, 1999; Fatoki and Mathabatha, 2001). The containers were then sealed in polyethylene water bottles. Ultrapure nitric acid was used to stabilise the water samples (0.5% HNO₃) (Mukherjee and Sirshendu, 2015; Momtaz, 2002; Chen et al., 2007). A quantitative method was used to accurately determine metal concentrations in waters, and so a multi-standard calibration method was used. Strong metal concentrations (Cd, Cr, Cu, Fe, Mn, Ni, Pb, and Zn) were measured by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-OES-Perki Elmer-Optima 4300 DV) at NGRI in Hyderabad,

Telangana (Truus et al., 2007; Kmiecik et al., 2016; Yang et al., 2016).

4. Results and discussion

The variations of the heavy metals (Figs. 2 & 3). Bottom water samples in Premonsoon (2016) and Postmonsoon (2017) from the thirty stations of the study area are summarized (Tables 1 & 2). The largest concentrations of sites are the Pb and Mn in both seasons at the River mouth of the area, LKVD -21 & 25 was recorded. This can be attributed to the natural sources of iron-rich laterites and the geological recent alluvium of the research area.

4.1. Level of heavy metals in seawater

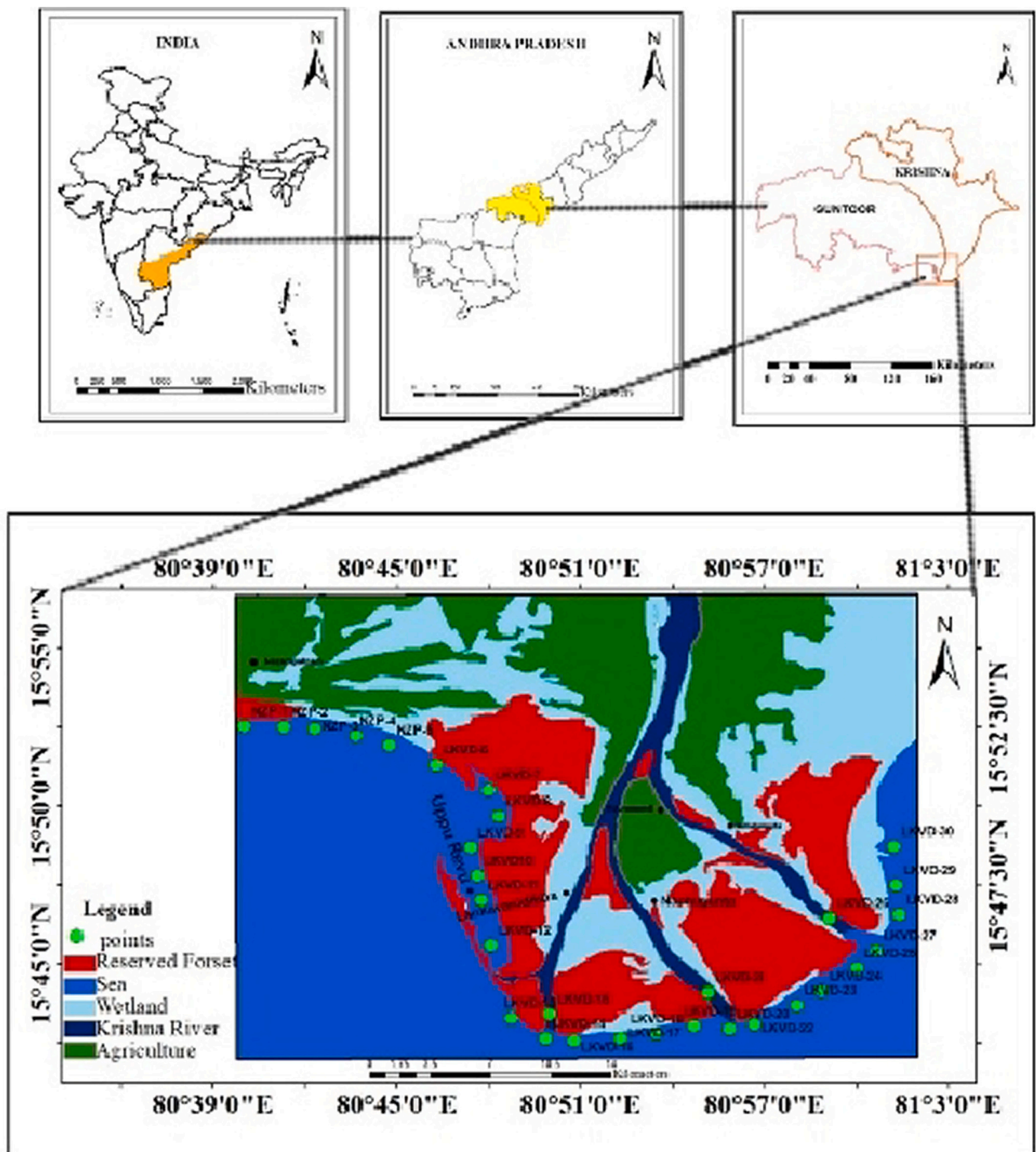
The level of heavy metals in seawater Pb & Mn shown in Tables 1 & 2. The mean concentrations of these heavy metals in seawater were in an ascending order: Pre monsoon: Pb (16.049 µg/L), > Mn (15.976 µg/L), > Cu (11.744 µg/L), > Cr (10.486 µg/L), > Fe (10.028 µg/L), > Cd (9.221 µg/L), > Ni (8.827 µg/L), > Zn (7.998 µg/L). Post monsoon: Mn (20.437 µg/L), > Pb (15.952 µg/L), > Cu (12.116 µg/L), > Ni (8.583 µg/L), > Zn (8.084 µg/L), > Cr (8.008 µg/L) > Cd (4.837 µg/L), > Fe (0.763 µg/L). The comparisons with the premonsoon and post monsoon (Tables 1 & 2) indicate that the average concentrations of Pb and Mn were the highest among all the locations. Meanwhile, the average concentration of Mn was higher than that in most locations of study area (Zhang et al., 2017). In general, Pb, Mn, and Cu mainly dominate metals pollution in the study area. The typically higher values of heavy metals in a majority of cases of land based anthropogenic sources (Table 3).

4.2. Hierarchical clustering

Here performed a hierarchical cluster analysis (HCA) on the study area. Three clusters were noticed in pre-monsoon (Fig. 4) Cluster 1 is made up of stations, LKVD-7, LKVD-6, NZP-3, NZP-2, LKVD-9, LKVD-11, NZP-5, LKVD-10, LKVD-18, and NZP-1 external effluent effect at both the survey region's inlets (Northern and Southern). This has resulted from its proximity to the stations. Cluster 2, constitutes with station LKVD-30, LKVD-17, LKVD-22, LKVD-19, and LKVD-23 show the higher values of heavy metals. In Cluster 3, LKVD-27, LKVD-25, LKVD-29, LKVD-8, LKVD-26, LKVD-24, LKVD-12, LKVD-15, LKVD-20, LKVD-28, LKVD-16, LKVD-13, and LKVD-14 showed mild contamination. Finally, In Post monsoon, three clusters were formed (Fig. 5) Cluster 1 is made up of stations LKVD-15, NZP-2, LKVD-25, LKVD-8, LKVD-16, LKVD-18, and NZP-1 of both heavy metals tested in the ecosystem of the coast. Cluster 2, made up of only LKVD -21 and NZP-4 stations, shows elevated levels of all heavy metals tested at the coastal opening on the south side of the research region. Though LKVD-22, LKVD-7, LKVD-11, NZP-5, LKVD-10, LKVD-9, LKVD-29, LKVD-19, 12, LKVD-17, LKVD-13, LKVD-30, LKVD-27, LKVD-26, LKVD-14, LKVD-28, LKVD-24, LKVD-6, LKVD-23, LKVD-20, and NZP-3 Cluster 3 formed, with the highest concentration.

4.3. Factor analysis (FA)

The data so obtained for various assessment of the coastal water of the heavy metal, were organized in a matrix with variables as columns and the samples as rows. Excel stat version 2018 was used for carrying out the statistical analysis of the data after performing auto scaling for various parameters. Factor analysis was performed on the heavy metal data that produced. During the pre-monsoon season, the study yielded three factors: Factor 1 (71.40%), Factor 2 (9.11%) and Factor 3 (4.38%) (Figs. 6 & 7). The Ni (0.887) and Cd (0.931) represented Factor 1. Hence it may be termed as Ni—Cd assemblage. It was determined that naming the assembly assemblages after their most important elements would be useful. Factor 2 is a Fe - Cd assemblage made of Fe (0.627) and Cd (0.351). Factor 3 is made up of Cu (0.105) and Cr (0.442), and it is known as the Cu—Cr assemblage (Figs. 10, 11, 12 & 13). According to



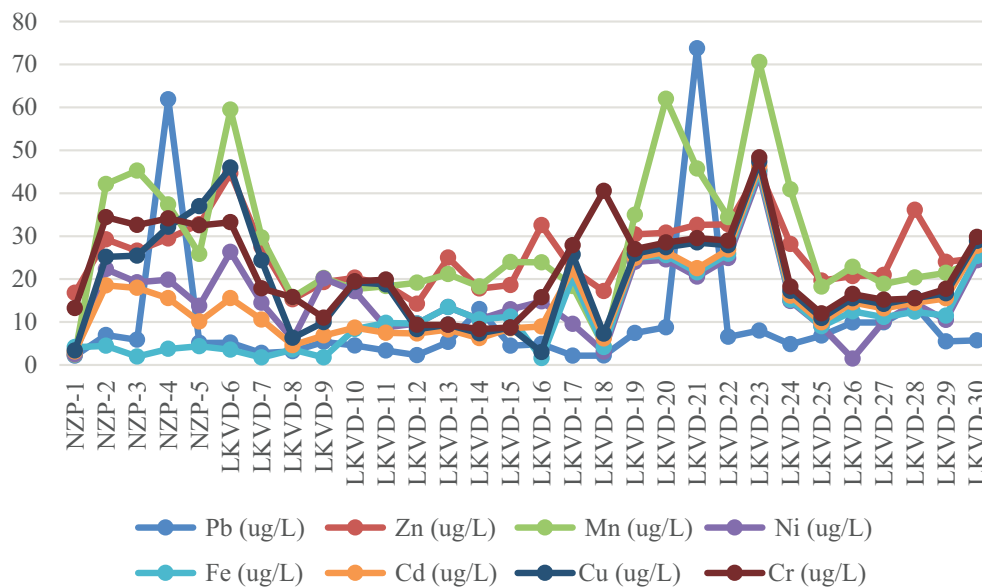


Fig. 2. Graphical representation of study area pre monsoon.

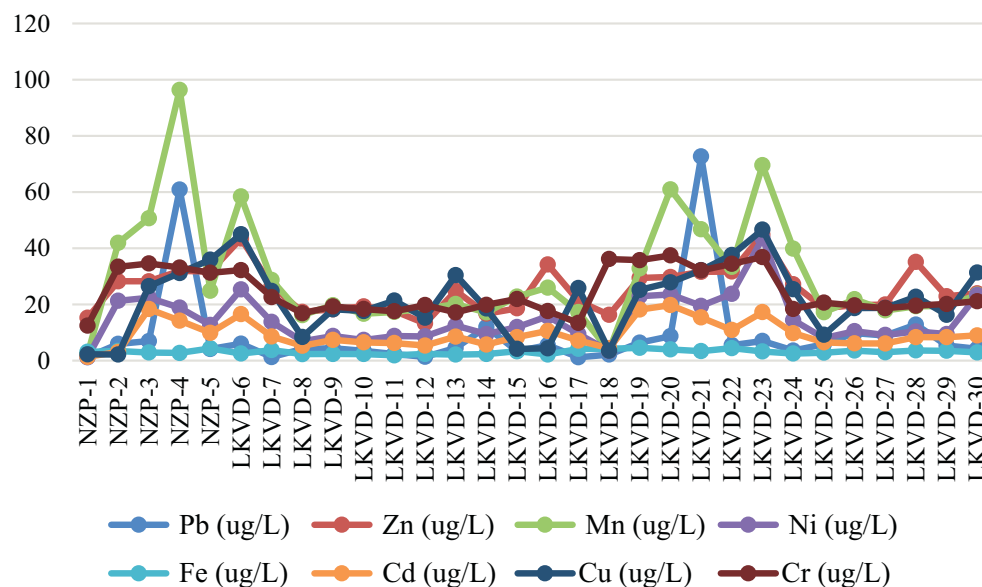


Fig. 3. Graphical representation of study area post monsoon.

how the different principal components were extracted. The eigenvalues for different factors, percentage variance accounted, cumulative percentage variance and component loadings (unrotated and Varimax rotated).

4.4. Pre monsoon and post monsoon

Thus, Biplots of PC1 vs PC6 were created as a result (Figs. 14, 15, 16 & 17). Which displayed samples from the sea level and sea floor, as well

as vector loadings. This diagram shows that nitrate-nitrogen, dissolved oxygen, and ammonia-cal-nitrogen are first quadrant, whereas salinity and ammonia-cal-nitrogen are in the third. Temperature, chlorophyll *a*, and phosphate are among the variables in the fourth quadrant. The presence of multiple variables in the same quadrant suggests a strong relationship. The link between dissolved oxygen, nitrite, and nitrate is well understood since their equilibrium values are determined by oxidation-reduction processes. The link between chlorophyll, temperature, and phosphate can be explained by assuming that phosphate

Table 1Heavy metal concentrations ($\mu\text{g/L}$) from the coastal waters of study area (Pre Monsoon).

Samples	Pb ($\mu\text{g/L}$)	Zn ($\mu\text{g/L}$)	Mn ($\mu\text{g/L}$)	Ni ($\mu\text{g/L}$)	Fe ($\mu\text{g/L}$)	Cd ($\mu\text{g/L}$)	Cu ($\mu\text{g/L}$)	Cr ($\mu\text{g/L}$)
NZP-1	2.13	16.81	2.37	2.47	4.13	3.11	3.34	13.25
NZP-2	6.95	29.25	42.14	22.3	4.46	18.53	25.17	34.41
NZP-3	5.85	26.59	45.26	19.25	1.95	17.95	25.45	32.59
NZP-4	61.88	29.46	37.38	19.85	3.74	15.54	32.17	34.14
NZP-5	5.14	32.76	25.84	13.76	4.35	10.11	36.96	32.52
LKVD-6	5.17	44.66	59.45	26.34	3.53	15.54	45.97	33.24
LKVD-7	2.77	27.52	29.69	14.47	1.74	10.57	24.34	17.83
LKVD-8	3.18	15.28	15.78	5.98	3.58	4.59	6.29	15.78
LKVD-9	5.38	19.29	20.24	20.13	1.76	6.85	9.97	10.99
LKVD-10	4.52	20.31	17.69	17.14	8.36	8.68	19.27	19.48
LKVD-11	3.35	18.68	18.34	8.78	9.78	7.47	18.77	19.87
LKVD-12	2.28	14.23	19.15	9.65	9.65	7.32	8.33	9.34
LKVD-13	5.33	24.99	21.15	13.46	13.46	8.24	9.38	9.35
LKVD-14	12.99	17.78	18.26	10.61	10.61	6.22	7.35	8.36
LKVD-15	4.47	18.61	23.97	12.95	11.25	8.52	8.68	8.72
LKVD-16	4.85	32.56	23.89	14.78	1.58	8.97	2.98	15.76
LKVD-17	2.15	22.21	18.34	9.54	20.45	22.24	25.72	27.88
LKVD-18	2.17	17.18	4.29	3.12	4.13	6.15	7.25	40.54
LKVD-19	7.44	30.42	34.96	23.98	24.98	24.9	25.92	26.98
LKVD-20	8.76	30.79	61.97	24.54	25.55	26.37	27.42	28.51
LKVD-21	73.74	32.61	45.74	20.54	21.57	22.51	28.52	29.53
LKVD-22	6.53	32.75	34.39	24.88	25.88	26.88	27.89	28.98
LKVD-23	7.99	45.27	70.56	44.33	45.34	46.31	47.32	48.32
LKVD-24	4.81	28.17	40.87	14.87	15.17	16.11	17.22	18.23
LKVD-25	6.82	19.65	18.22	8.99	8.98	9.91	10.92	11.98
LKVD-26	9.88	20.69	22.84	1.47	12.48	14.56	15.51	16.51
LKVD-27	9.91	20.96	18.87	10.14	11.15	13.17	14.18	15.19
LKVD-28	13.83	36.14	20.34	14.37	12.38	14.49	15.58	15.51
LKVD-29	5.47	23.99	21.43	10.47	11.48	15.51	16.67	17.71
LKVD-30	5.71	24.91	24.45	24.34	25.35	27.71	28.89	29.81
Minimum	2.13	14.23	2.37	1.47	1.58	3.11	2.98	8.36
Maximum	73.74	45.27	70.56	44.33	45.34	46.31	47.32	48.32
Mean	10.048	25.817	28.596	15.583	11.961	14.834	19.781	22.377
Std. deviation	16.049	7.998	15.976	8.827	10.028	9.221	11.744	10.486

Table 2Heavy metal concentrations ($\mu\text{g/L}$) from the coastal waters of study area (Post Monsoon).

Samples	Pb ($\mu\text{g/L}$)	Zn ($\mu\text{g/L}$)	Mn ($\mu\text{g/L}$)	Ni ($\mu\text{g/L}$)	Fe ($\mu\text{g/L}$)	Cd ($\mu\text{g/L}$)	Cu ($\mu\text{g/L}$)	Cr ($\mu\text{g/L}$)
NZP-1	1.25	15.27	2.24	1.16	3.12	1.34	2.25	12.46
NZP-2	5.94	28.24	41.9	21.29	3.45	3.11	2.16	33.41
NZP-3	6.96	28.25	50.69	22.35	2.88	18.37	26.48	34.55
NZP-4	60.87	31.95	96.37	18.84	2.73	14.23	31.16	33.13
NZP-5	4.15	31.76	24.83	12.66	4.34	9.88	35.95	31.25
LKVD-6	5.99	43.45	58.44	25.33	2.52	16.53	44.96	32.22
LKVD-7	1.26	26.51	28.68	13.76	3.73	8.56	24.78	22.58
LKVD-8	4.17	17.36	16.19	6.88	2.32	5.27	8.38	16.83
LKVD-9	4.37	18.28	19.73	8.77	2.38	7.37	18.27	19.27
LKVD-10	3.51	19.32	16.68	7.35	2.33	6.48	17.66	18.36
LKVD-11	2.35	17.67	17.33	8.77	1.89	6.25	21.37	17.56
LKVD-12	1.37	13.22	18.44	8.64	2.37	5.35	15.23	19.75
LKVD-13	4.52	24.83	20.14	12.45	2.16	8.54	30.43	17.16
LKVD-14	11.98	16.72	17.15	9.46	2.34	5.73	18.54	19.87
LKVD-15	3.38	18.59	22.8	11.94	3.35	8.37	4.17	21.93
LKVD-16	5.53	34.21	25.96	15.77	2.19	10.56	4.38	17.61
LKVD-17	1.18	21.19	17.25	9.46	4.17	7.15	25.81	13.41
LKVD-18	2.16	16.27	3.48	3.41	4.21	4.42	3.53	36.15
LKVD-19	6.39	29.42	32.78	22.92	4.54	18.17	25.17	35.76
LKVD-20	8.69	29.76	60.96	23.54	3.99	19.75	27.84	37.43
LKVD-21	72.68	31.6	46.73	19.48	3.37	15.35	32.17	32.11
LKVD-22	5.52	31.68	33.38	23.77	4.44	10.89	37.59	34.44
LKVD-23	6.98	44.87	69.55	43.55	3.26	17.26	46.61	36.87
LKVD-24	3.58	27.16	39.86	14.42	2.52	9.77	25.39	18.38
LKVD-25	5.81	18.64	17.21	7.99	2.83	6.49	9.17	20.63
LKVD-26	9.77	19.77	21.83	10.47	3.57	6.14	18.68	19.64
LKVD-27	8.92	19.95	17.86	9.13	2.98	6.11	18.77	18.65
LKVD-28	12.83	35.13	19.33	10.37	3.63	8.37	22.73	19.54
LKVD-29	5.35	22.91	19.42	9.47	3.51	8.32	16.32	20.15
LKVD-30	4.21	23.92	23.42	23.34	2.92	8.96	31.34	21.14
Minimum	1.18	13.22	2.24	1.16	1.89	1.34	2.16	12.46
Maximum	72.68	44.87	96.37	43.55	4.54	19.75	46.61	37.43
Mean	9.389	25.263	30.021	14.558	3.135	9.436	21.576	24.408
Std. deviation	15.952	8.084	20.437	8.583	0.763	4.837	12.116	8.008

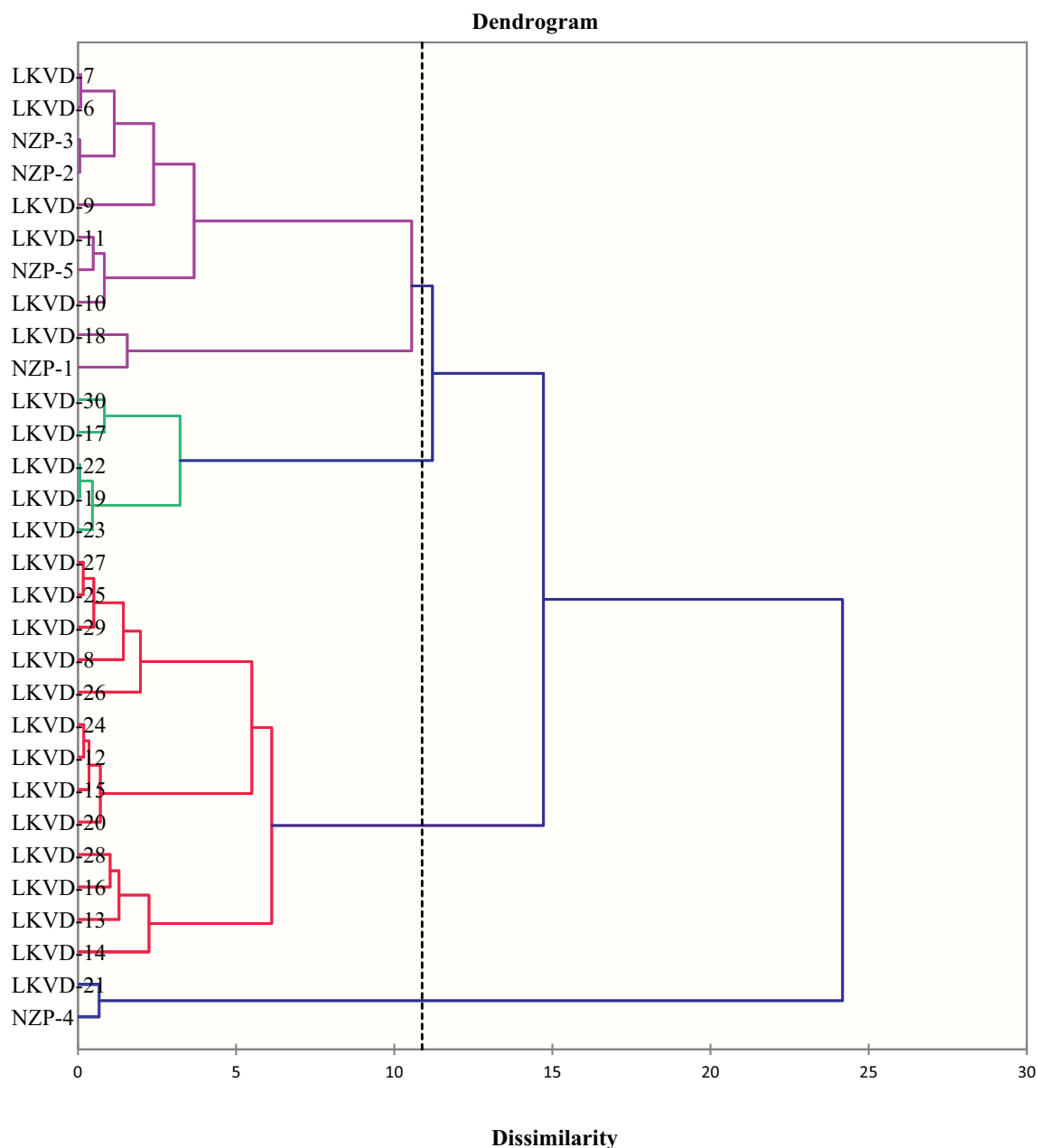
Table 3

Rainfall (mm) data - Year 2016 & 2017.

District-wise seasonal and annual Rainfall (mm) - Year 2017				
Districts	Pre- Monsoon	Monsoon	Post- Monsoon	Annual
Guntur (Nizampatnam)	49.5	640.9	101.2	791.6
Krishna (Lankavanidibba)	33.9	706	127.8	867.7

District-Wise Seasonal And Annual Rainfall (Mm) - Year 2016				
Districts	Pre-Monsoon	Monsoon	Post-Monsoon	Annual
Guntur (Nizampatnam)	113	771.1	81	965.1
Krishna (Lankavanidibba)	119.9	801	95	1015.9

(Rainfall Statistics of India – 2016 & 2017) (49 & 52).

**Fig. 4.** Dendrogram of grouping of bottom water sampling stations during pre-monsoon.

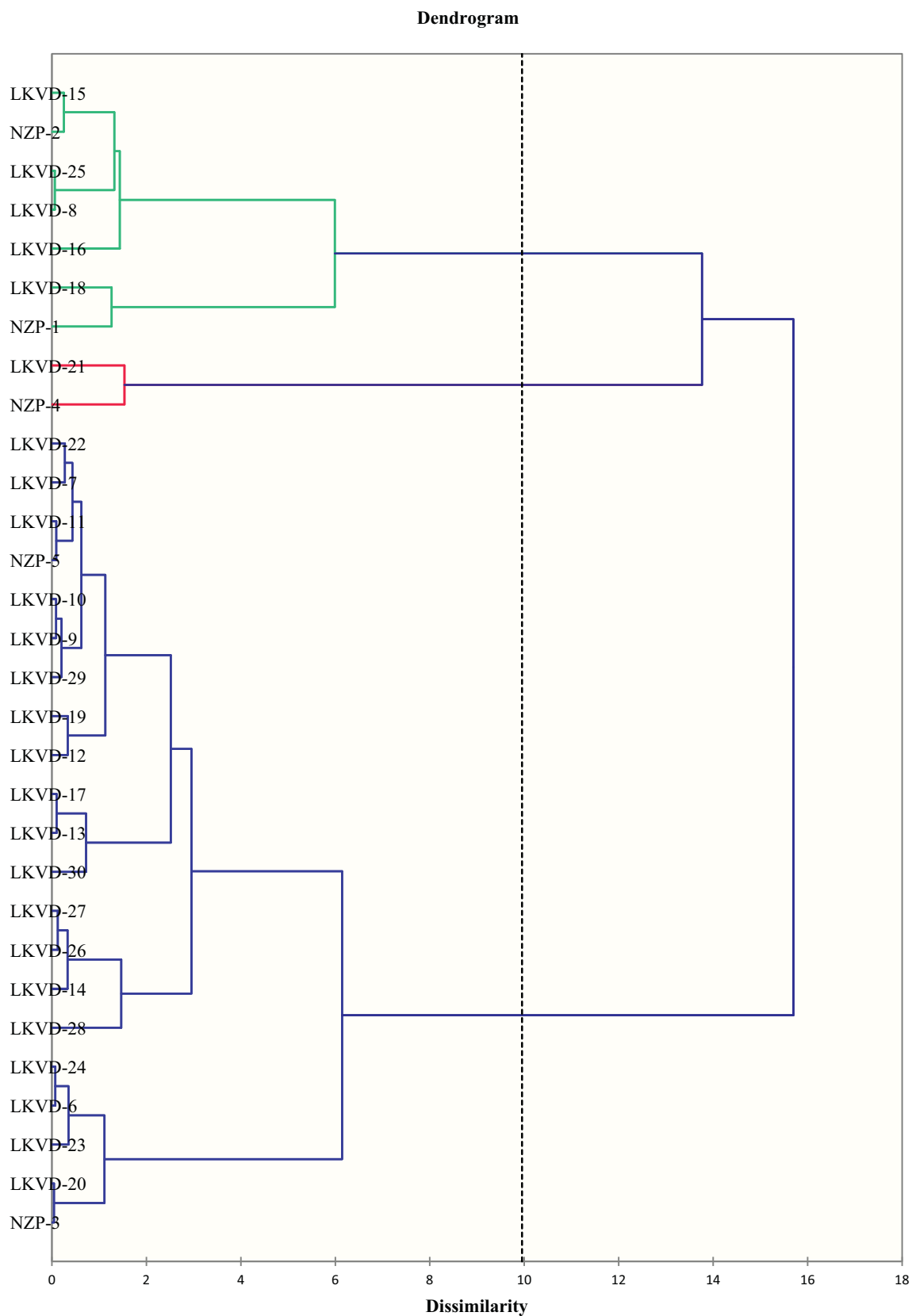


Fig. 5. Dendrogram of grouping of bottom water sampling stations during post monsoon.

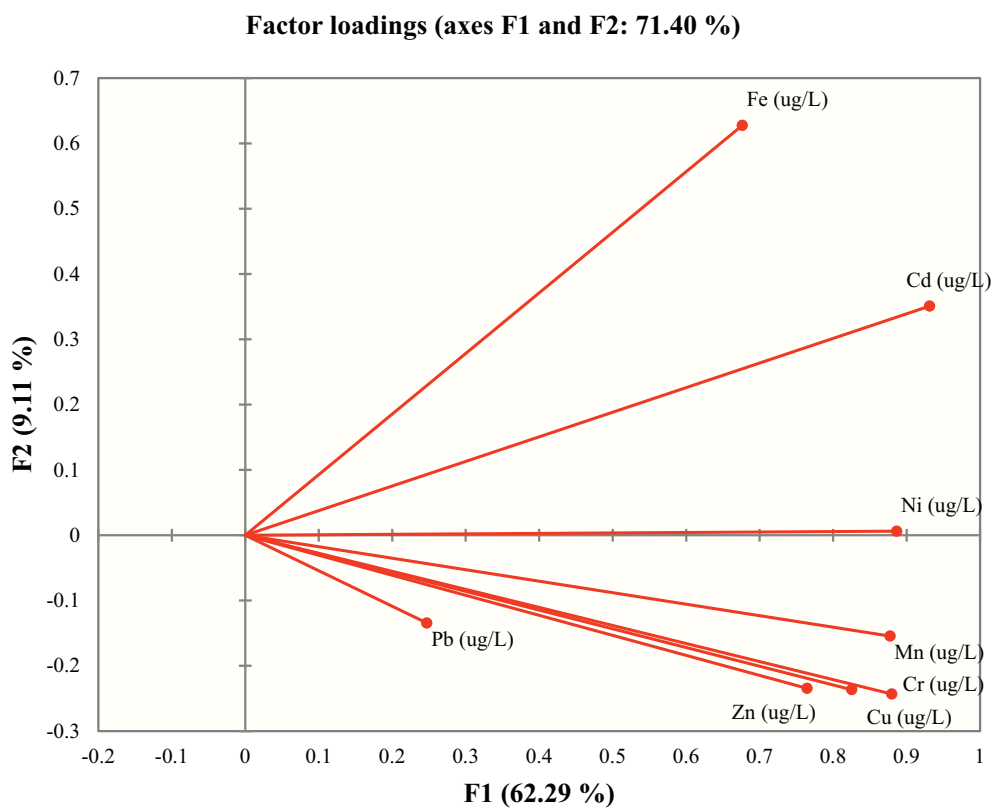


Fig. 6. Factor loading during pre-monsoon (F1& F2).

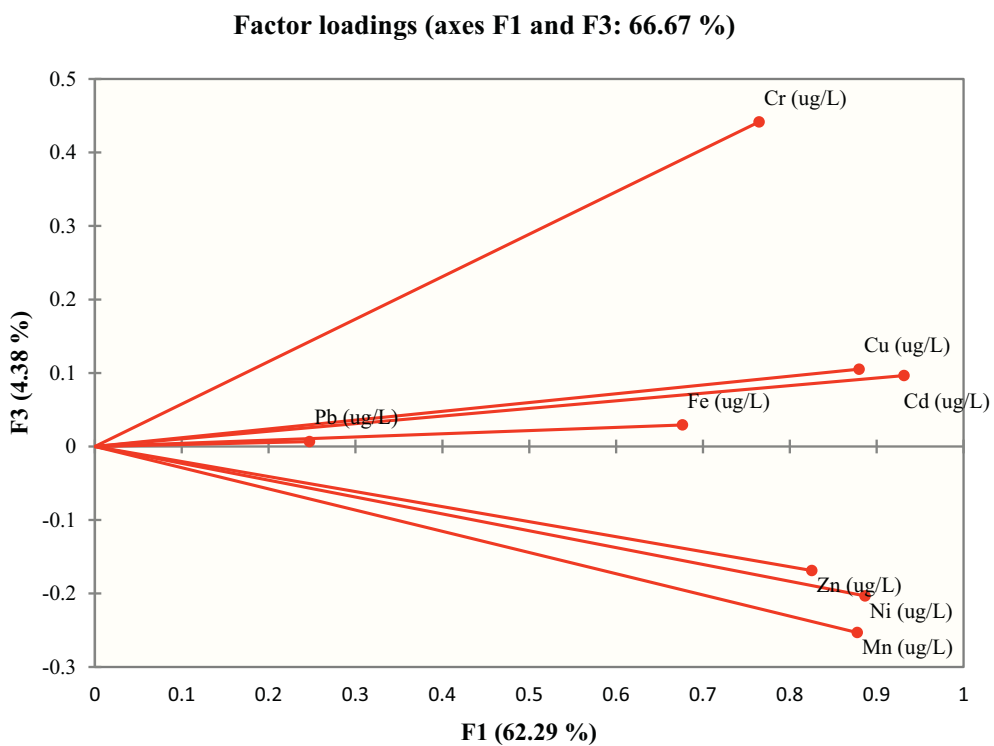


Fig. 7. Factor loading during pre-monsoon (F1&F3).

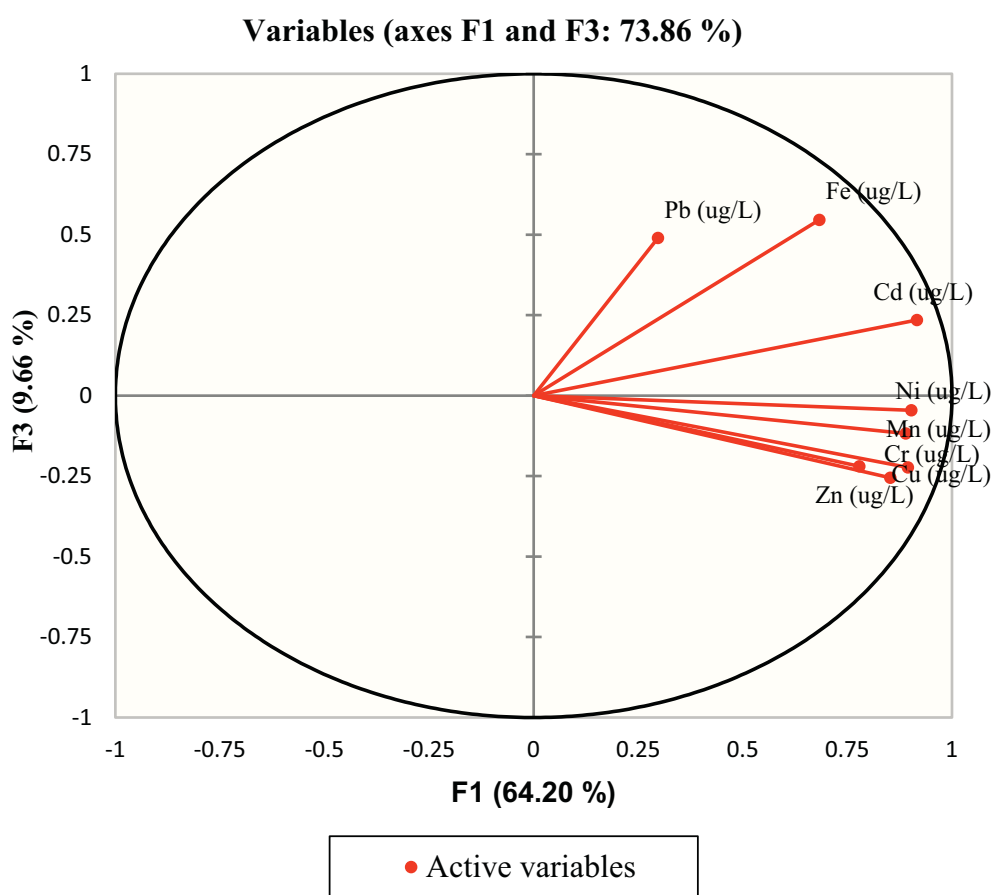


Fig. 8. Variables (PCA) during pre-monsoon (F1&F3).

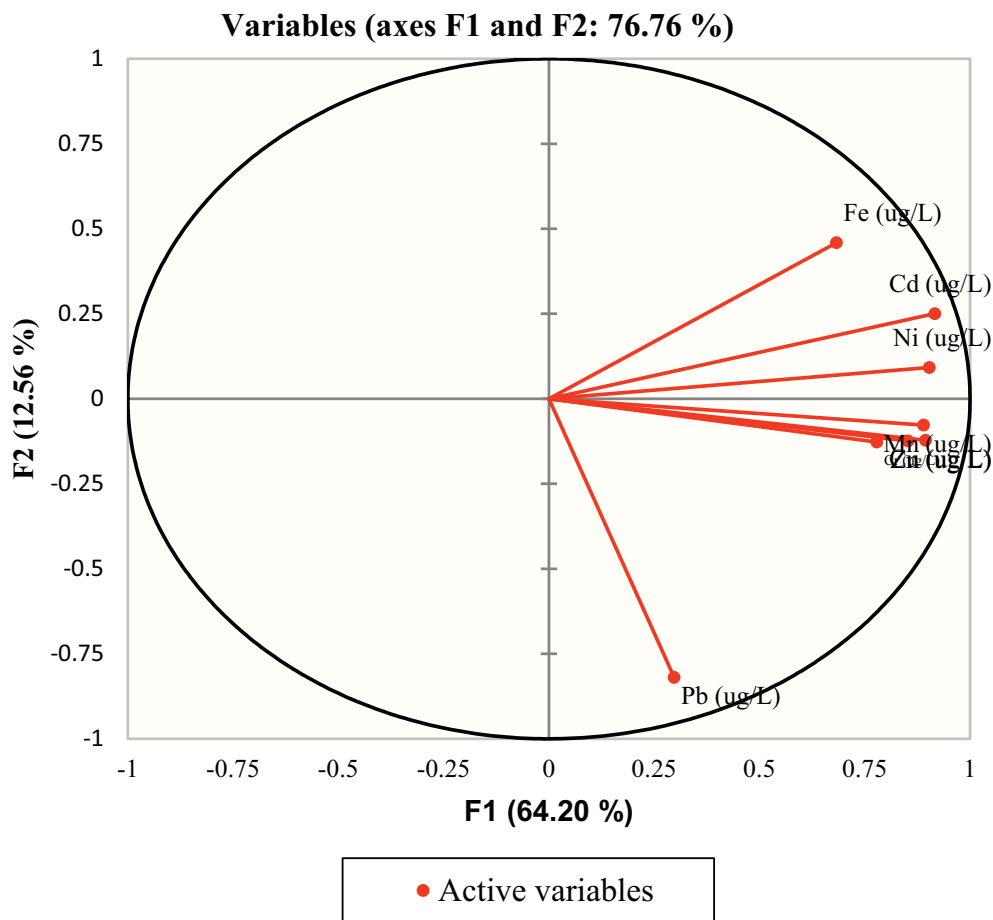


Fig. 9. Variables (PCA) during pre-monsoon (F1&F2).

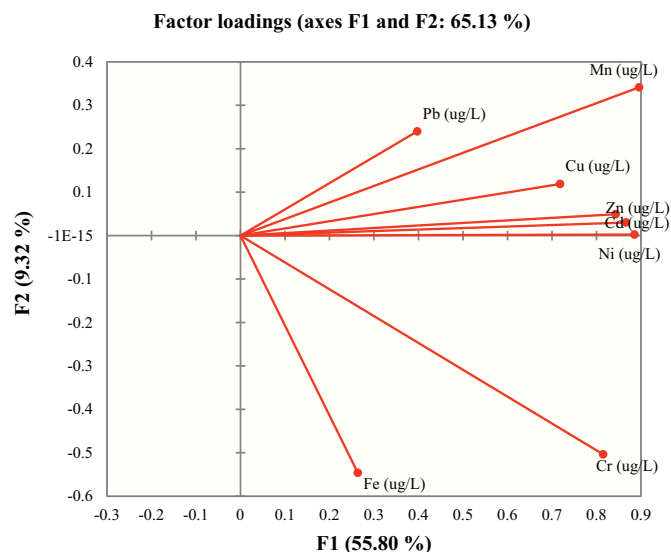


Fig. 10. Factor loading during Pre monsoon (F1&F2).

increases aquatic plant development, which in turn promotes chlorophyll growth. Temperature also has a vital role in photosynthesis. The presence of ammonia and salinity in the third quadrant shows an ammonia influx in coastal waters caused by Krishna River discharges, which carry untreated effluents from nearby industry. Freshwater influx

causes salinity dilution. When looking at the map, one can notice that phosphate and ammonia are in distinct quadrants. As a result, they are self-sufficient. As a result, the origins of ammonia and phosphate cannot be the same. It has been determined that there is extensive hauling activity of rock-phosphate in the vicinity, which is a completely different source than ammonia. In there are two clusters of points: one in the third quadrant, which displays ammonia and salinity, and one in the fourth quadrant, which shows temperature and salinity. The other is in the fourth quadrant, which includes temperature, chlorophyll *a*, and phosphate. The first cluster occurs between February and April, while the second between November and December. It is worth noting that the former occurs during the summer months, whereas the latter occurs throughout the winter months.

4.5. Factor score

The pre-monsoon factor score is highest in F1, F2 and F3 and Post monsoon also the same (Table 4). F1, F2 and F3 recorded highest score at LKVD-23 (2.289), LKVD-22 (4.373), and LKVD-16 (3.098) and also post-monsoon (Table 5). F1, F2 and F3 highest score at LKVD-23 (2.454), LKVD-7 (0.275) and LKVD-16 (0.466).

4.6. Pearson's correlation matrix (PCM)

The relationship among metal concentrations was examined by PCM (Tables 6 & 7). Further studies demonstrated positive correlations with pre monsoon, Ni and Cr metal concentrations in the organs and blood demonstrated common routes of distribution, and mutually supportive relationship, and organ toxicity pointed to the route of metal transport.

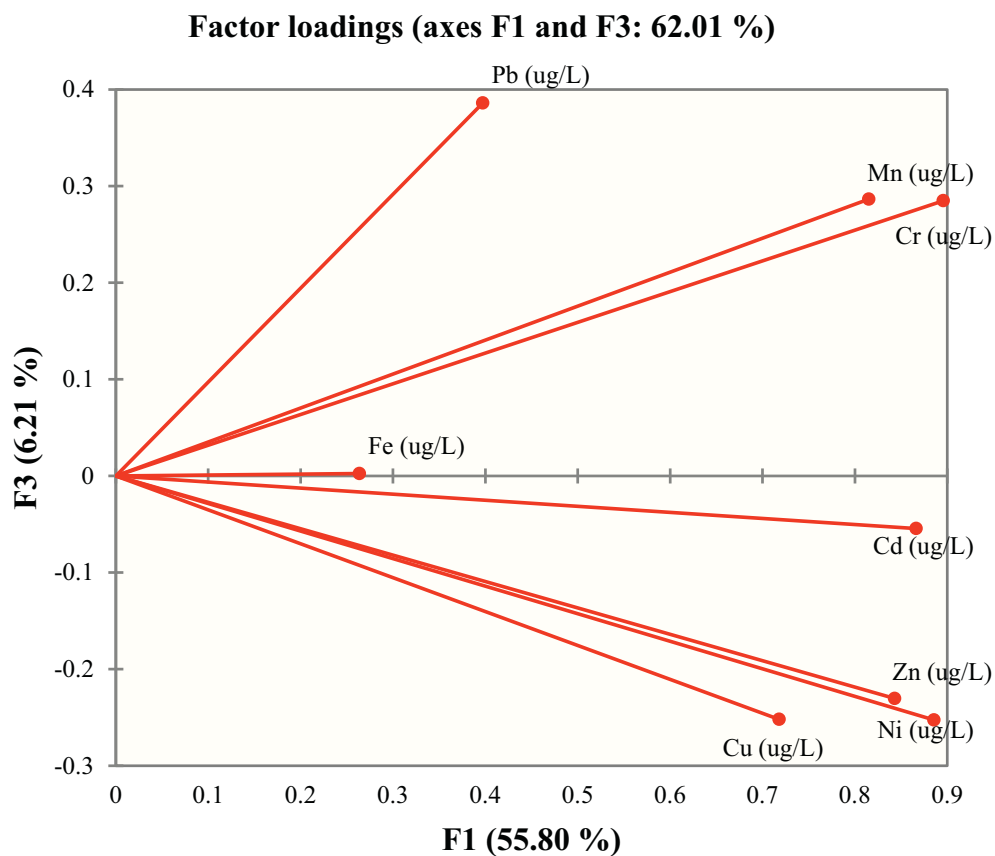


Fig. 11. Factor loading during Pre monsoon (F1&F3).

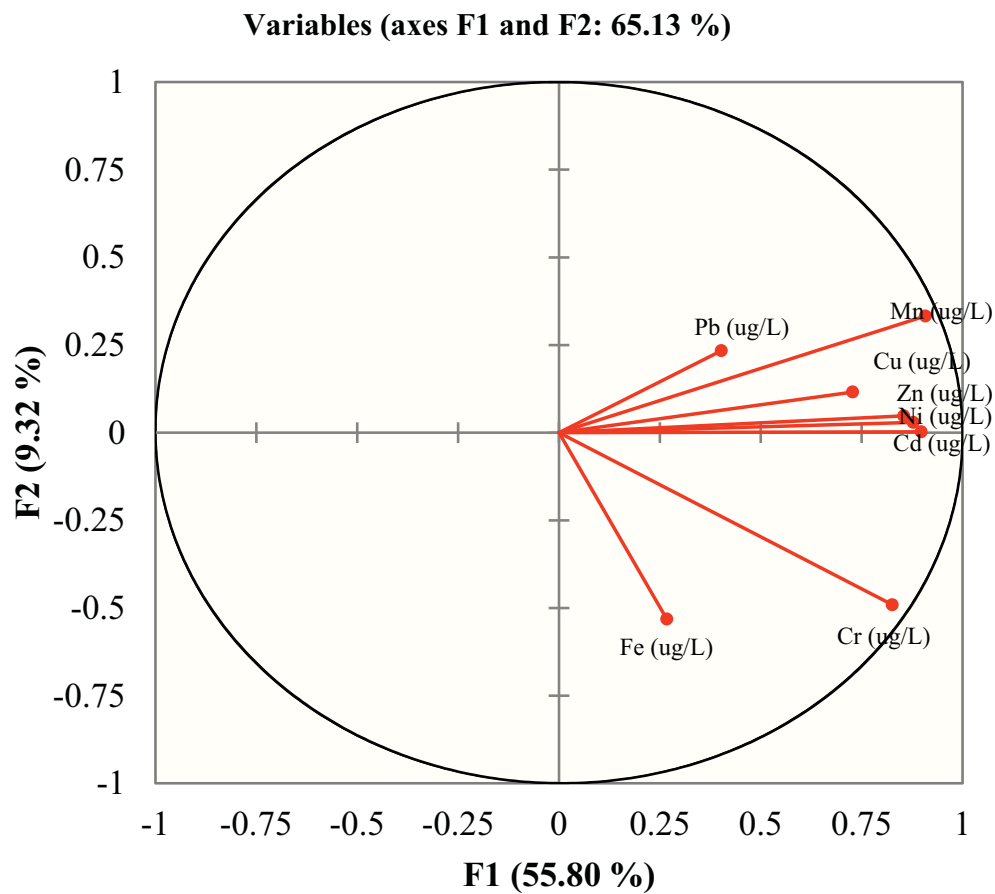


Fig. 12. Variables (PCA) during post monsoon (F1&F2).

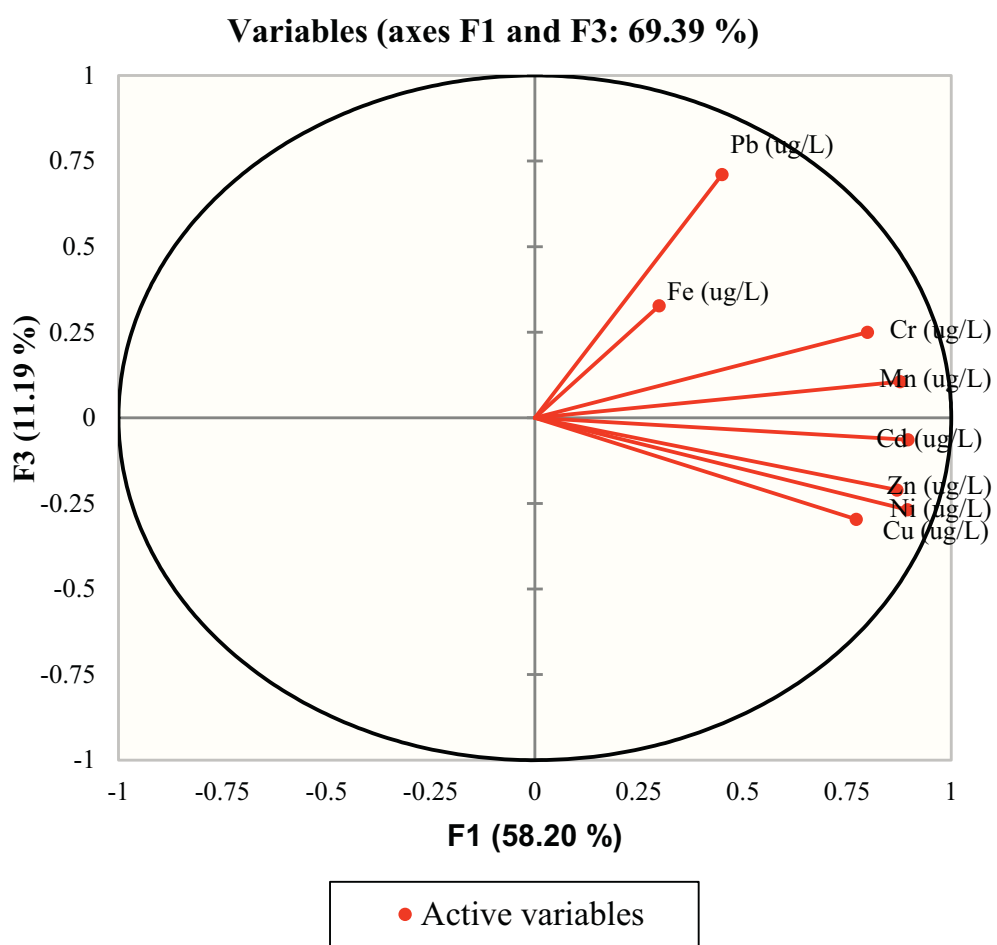


Fig. 13. Variables (PCA) during post monsoon (F1& F3).

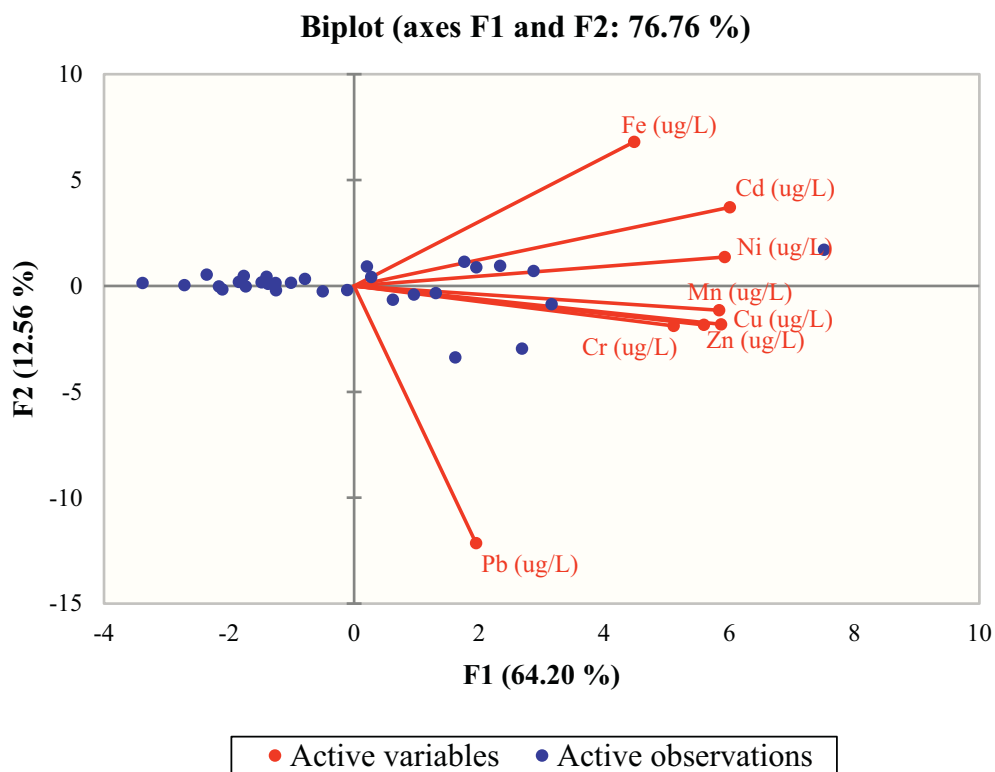


Fig. 14. Biplot pre monsoon (F1&F2).

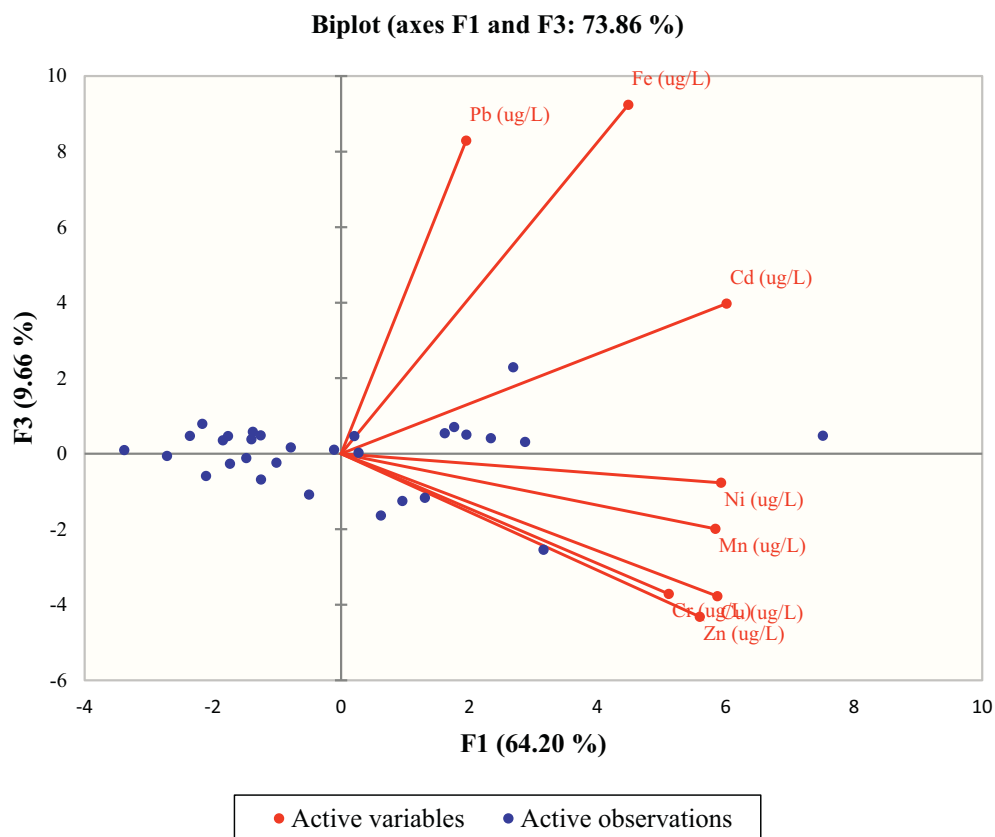


Fig. 15. Biplot premonsoon (F1&F3).

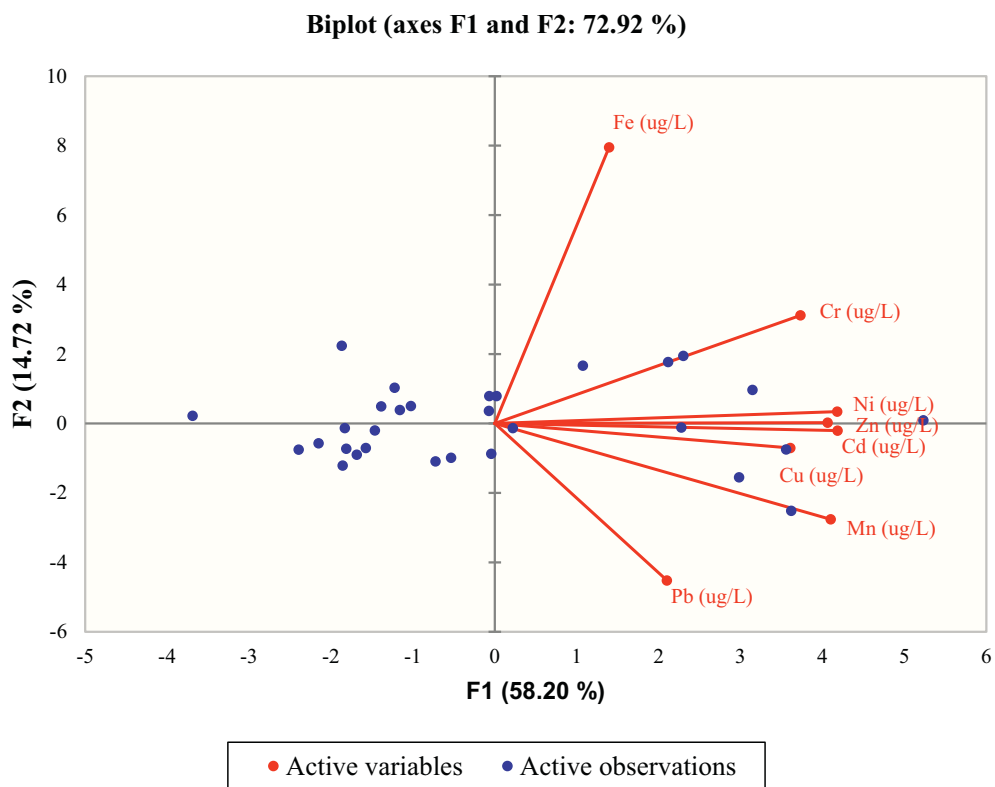


Fig. 16. Biplot post monsoon (F1&F2).

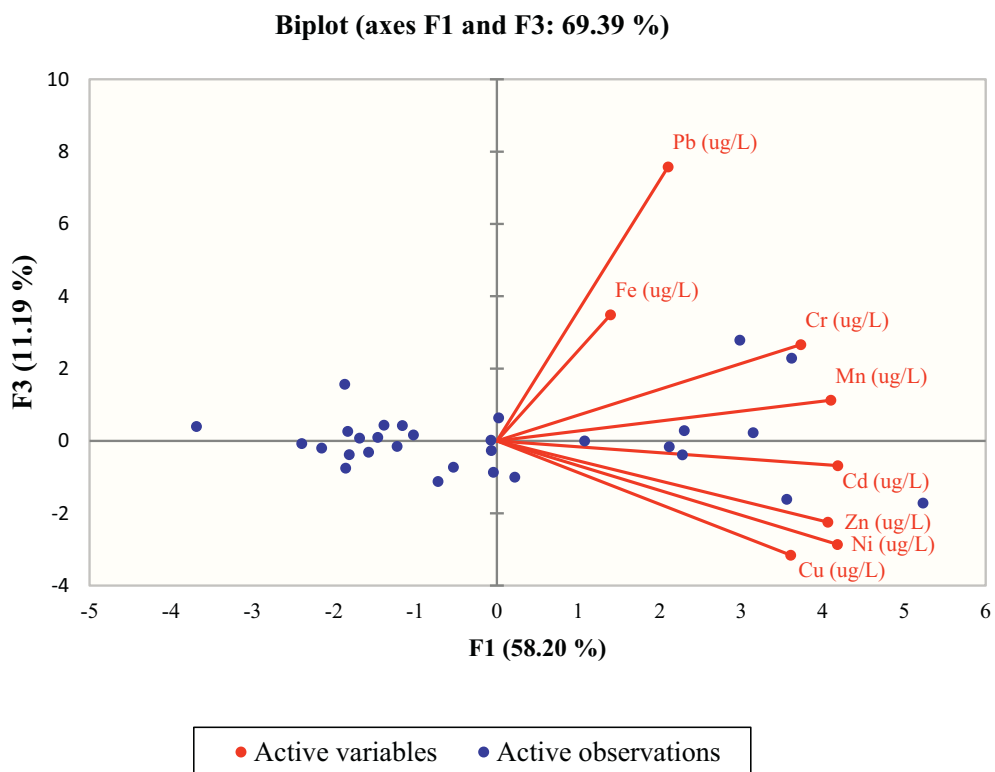


Fig. 17. Biplot post monsoon (F1&F2).

Table 4
Factor score during pre-monsoon.

Observation	F1	F2	F3
NZP-1	−1.506	0.167	0.451
NZP-2	0.718	−0.871	0.355
NZP-3	0.657	−0.896	0.352
NZP-4	0.602	−1.425	0.593
NZP-5	0.119	−1.942	0.789
LKVD-6	1.248	−2.819	−1.025
LKVD-7	−0.114	−0.820	−0.401
LKVD-8	−1.193	−0.159	0.121
LKVD-9	−0.723	−0.165	−1.001
LKVD-10	−0.540	−0.370	−0.029
LKVD-11	−0.761	−0.384	0.327
LKVD-12	−1.020	0.501	−0.532
LKVD-13	−0.777	0.266	−1.158
LKVD-14	−1.064	0.339	−0.838
LKVD-15	−0.812	0.436	−1.028
LKVD-16	−0.527	−0.308	−0.934
LKVD-17	0.279	1.212	1.675
LKVD-18	−1.071	−0.834	2.573
LKVD-19	0.855	0.916	0.032
LKVD-20	1.244	0.352	−0.885
LKVD-21	0.924	−0.041	−0.140
LKVD-22	1.048	0.993	0.227
LKVD-23	3.246	1.200	0.019
LKVD-24	0.122	0.166	−0.861
LKVD-25	−0.739	0.504	−0.213
LKVD-26	−0.420	0.813	0.598
LKVD-27	−0.446	0.687	0.127
LKVD-28	−0.086	0.297	−0.537
LKVD-29	−0.184	0.675	0.280
LKVD-30	0.924	1.509	1.064

Table 5
Factor score during post monsoon.

Observation	F1	F2	F3
NZP-1	−1.689	0.365	0.052
NZP-2	0.336	−0.825	0.990
NZP-3	1.162	−0.360	0.475
NZP-4	1.825	2.556	3.237
NZP-5	0.428	−1.467	0.135
LKVD-6	1.712	0.176	−0.274
LKVD-7	−0.033	0.289	−0.173
LKVD-8	−1.028	0.409	0.209
LKVD-9	−0.684	0.218	0.167
LKVD-10	−0.798	0.144	0.111
LKVD-11	−0.805	0.279	−0.019
LKVD-12	−0.876	0.115	0.498
LKVD-13	−0.413	0.456	−0.860
LKVD-14	−0.787	−0.079	0.117
LKVD-15	−0.573	0.139	0.145
LKVD-16	−0.253	0.703	−1.229
LKVD-17	−0.731	1.249	−0.853
LKVD-18	−0.717	−3.454	1.702
LKVD-19	0.933	−1.520	−0.627
LKVD-20	1.568	−0.049	0.939
LKVD-21	0.921	−0.412	0.134
LKVD-22	0.895	−1.445	−0.578
LKVD-23	2.420	0.168	−1.696
LKVD-24	0.059	1.570	−0.038
LKVD-25	−0.792	−0.117	0.309
LKVD-26	−0.582	0.395	0.014
LKVD-27	−0.718	0.226	−0.094
LKVD-28	−0.221	0.028	−0.896
LKVD-29	−0.521	0.144	−0.159
LKVD-30	−0.040	0.101	−1.736

Table 6

Pearson correlation matrix for the selected heavy metals during pre-monsoon.

Variables	Pb (µg/L)	Zn (µg/L)	Mn (µg/L)	Ni (µg/L)	Fe (µg/L)	Cd (µg/L)	Cu (µg/L)	Cr (µg/L)
Pb (µg/L)	1							
Zn (µg/L)	0.234	1						
Mn (µg/L)	0.266	0.791	1					
Ni (µg/L)	0.180	0.772	0.835	1				
Fe (µg/L)	0.093	0.402	0.484	0.597	1			
Cd (µg/L)	0.181	0.672	0.746	0.805	0.855	1		
Cu (µg/L)	0.253	0.777	0.779	0.751	0.454	0.738	1	
Cr (µg/L)	0.223	0.605	0.593	0.596	0.377	0.678	0.776	1

Table 7

Pearson correlation matrix for the selected heavy metals during post monsoon.

Variables	Pb (µg/L)	Zn (µg/L)	Mn (µg/L)	Ni (µg/L)	Fe (µg/L)	Cd (µg/L)	Cu (µg/L)	Cr (µg/L)
Pb (µg/L)	1							
Zn (µg/L)	0.274	1						
Mn (µg/L)	0.550	0.711	1					
Ni (µg/L)	0.191	0.809	0.759	1				
Fe (µg/L)	−0.008	0.219	0.031	0.190	1			
Cd (µg/L)	0.348	0.728	0.758	0.768	0.215	1		
Cu (µg/L)	0.255	0.679	0.585	0.678	0.152	0.680	1	
Cr (µg/L)	0.308	0.584	0.654	0.684	0.497	0.680	0.427	1

Table 8

Factor loading matrix for heavy metals from pre monsoon.

Heavy M.	F1	F2	F3
Pb (µg/L)	0.247	−0.134	0.007
Zn (µg/L)	0.825	−0.237	−0.169
Mn (µg/L)	0.878	−0.154	−0.253
Ni (µg/L)	0.887	0.006	−0.203
Fe (µg/L)	0.676	0.627	0.029
Cd (µg/L)	0.931	0.351	0.096
Cu (µg/L)	0.880	−0.243	0.105
Cr (µg/L)	0.765	−0.234	0.442

Table 9

Factor loading matrix for heavy metals from post monsoon.

Heavy M.	F1	F2	F3
Pb (µg/L)	0.397	0.240	0.386
Zn (µg/L)	0.843	0.049	−0.230
Mn (µg/L)	0.896	0.341	0.285
Ni (µg/L)	0.885	0.002	−0.253
Fe (µg/L)	0.264	− 0.546	0.003
Cd (µg/L)	0.866	0.030	−0.054
Cu (µg/L)	0.718	0.119	−0.252
Cr (µg/L)	0.815	−0.504	0.287

These correlation results are key to understanding the relationships among metal concentrations.

Factor analysis was used on the heavy metal data that was analysed. During the Premonsoon, the study revealed three factors, specifically, Factor 1 (53.93%), Factor 2 (12.52%), and Factor 3 (3.45%). The Mn (0.912) represented Factor 1 (Tables 8 & 9). As a result, it is often referred to as a Co—Cr assemblage. The names of the assemblages were inspired by the most well-known heavy metal bands in their respective genres. Factor 2 was called Cr—Ni assemblage because it was made up of Cu (0.330) and Fe (0.902). Fe (0.295) and Pb (0.826) both expressed factor 3, which are known as the Fe—Pb assemblage. Factor loadings for the pre-monsoon season reveal three distinct factors: Factor 1 (60.26%), Factor 2 (13.12%), and Factor 3. (10.29%). Factor 1 has two variables: Ni (0.899) and Co. (0.909). As a result, it may be referred to as a Ni—Co assemblage. Factor 2 consists of two variables: Cr (0.366) and Fe (0.366). (0.872). As a result, it is referred to as Cr—Fe assemblage. Factor 3 is known as the Fe—Pb assemblage due to the presence of two

variables, Fe (0.213) and Pb (0.789). For both seasons, a cross plot of Factors 1&2 and Factors 1&3 provided significant data on heavy metal distribution as well as unique clusters of a single factor. Heavy metals, according to the distribution, have a strong affinity for the aquatic environments of Nizampatnam Bay and the Lankavanidibba coast on India's southeast coast.

5. Conclusion

Due to high effluent inflow from several unidentified sources, the effect of waste discharged in to the Bay of Bengal has been important in recent years. Heavy metal concentrations in Bay coastalwater were found to be relatively high and varied across sampling points. The combined use of different approaches for interpreting sediment characteristics in terms of context factors is needed for evaluating coastal water metal contamination. The findings suggest that particular attention should be paid to the problem of element remobilization, which leads to the release of a significant portion of elements in sediments and, eventually, back into coastal water. As a result, consistent monitoring of water quality on a regular basis is needed to detect any potential changes in quality and to reduce the spread of health disorders and their negative effects on the aquatic environment and human health. The evaluation of heavy metals in the from coastal water at Nizampatnam Bay and Lankavanidibba, East Coast of India was undertaken, whereby sand, mud, organic matter, and heavy metals such as Fe, Al, Mn, Cr, Cu, Ni, Cd, Pb, and Zn were analysed. The mean concentration of heavy metals was shown to decrease in the following order: Pre monsoon: Pb (16.049 µg/L), >Mn (15.976 µg/L), >Cu (11.744 µg/L), >Cr (10.486 µg/L), >Fe (10.028 µg/L), >Cd (9.221 µg/L), >Ni (8.827 µg/L), >Zn (7.998 µg/L). Post monsoon: Mn (20.437 µg/L), >Pb (15.952 µg/L), >Cu (12.116 µg/L), >Ni (8.583 µg/L), >Zn (8.084 µg/L), >Cr (8.008 µg/L) > Cd (4.837 µg/L), >Fe (0.763 µg/L). The coastal water observed in most of the stations was muddy water due to the shallow depth and poor wave action in the study area. The pollution indices such as Hierarchical clustering, Factor Analysis (FA), PCA, Biplots, Factor score, Person's Correlation Matrix (PCM), Factor loading matrix indicated that Pb, Mn was responsible for the high contamination level in the study area. The PCA results also confirmed that Pb, Mn had a high contamination level, as indicated by the strong positive loadings. The main source of Pb, Mn was due to anthropogenic inputs such as municipal wastewater, domestic sewage discharge, fishing harbour activities, and industrial and

aquaculture wastes. Based on the results obtained for Hierarchical clustering, Factor Analysis (FA), PCA, Biplots, Factor score, Person's Correlation Matrix (PCM), Factor loading matrix, it was found that the presence of heavy metals such as Cu, Zn, Pb, and Cr led to moderate contamination in the study area. The PCA results suggest that the coastal water in the study area were polluted by heavy metals. The findings of this study revealed that the study area frequently receives heavy metal contaminants from different sources and if the concentration of these heavy metals continues to increase, the toxicity will also increase, thus affecting the entire food chain within the marine ecosystem. Therefore, to protect the marine ecosystem, illegal discharges into the marine environment should be properly monitored and effluents from the industries, municipal, and domestic areas should be pretreated before its discharge into the coastal areas.

Declaration of Competing Interest

None.

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